

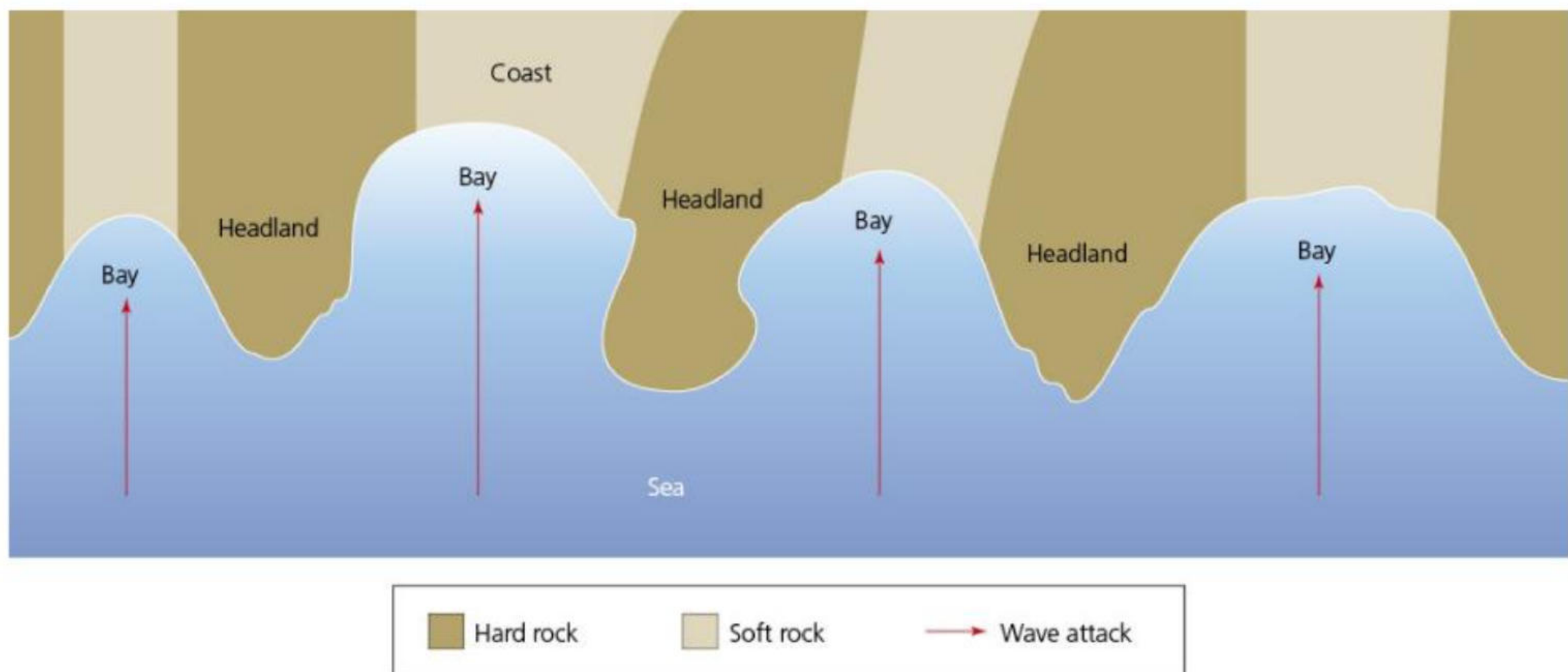
2.2 The main landforms associated with these processes

This chapter will explain:

- ★ the main landforms associated with coastal processes
- ★ the formation and main characteristics of discordant and concordant coastlines.

Headlands and bays and their features

Hard rocks form headlands that protrude, whereas weaker rocks are eroded to form **bays** (Figures 2.9 and 2.10). Wave refraction in the bay spreads wave energy around the bay, whereas it focuses wave energy on the flanks of the headlands. Bayhead beaches are formed when constructive waves deposit sand between two headlands, such as at Maracas Bay and Tyrico Bay in northern Trinidad.

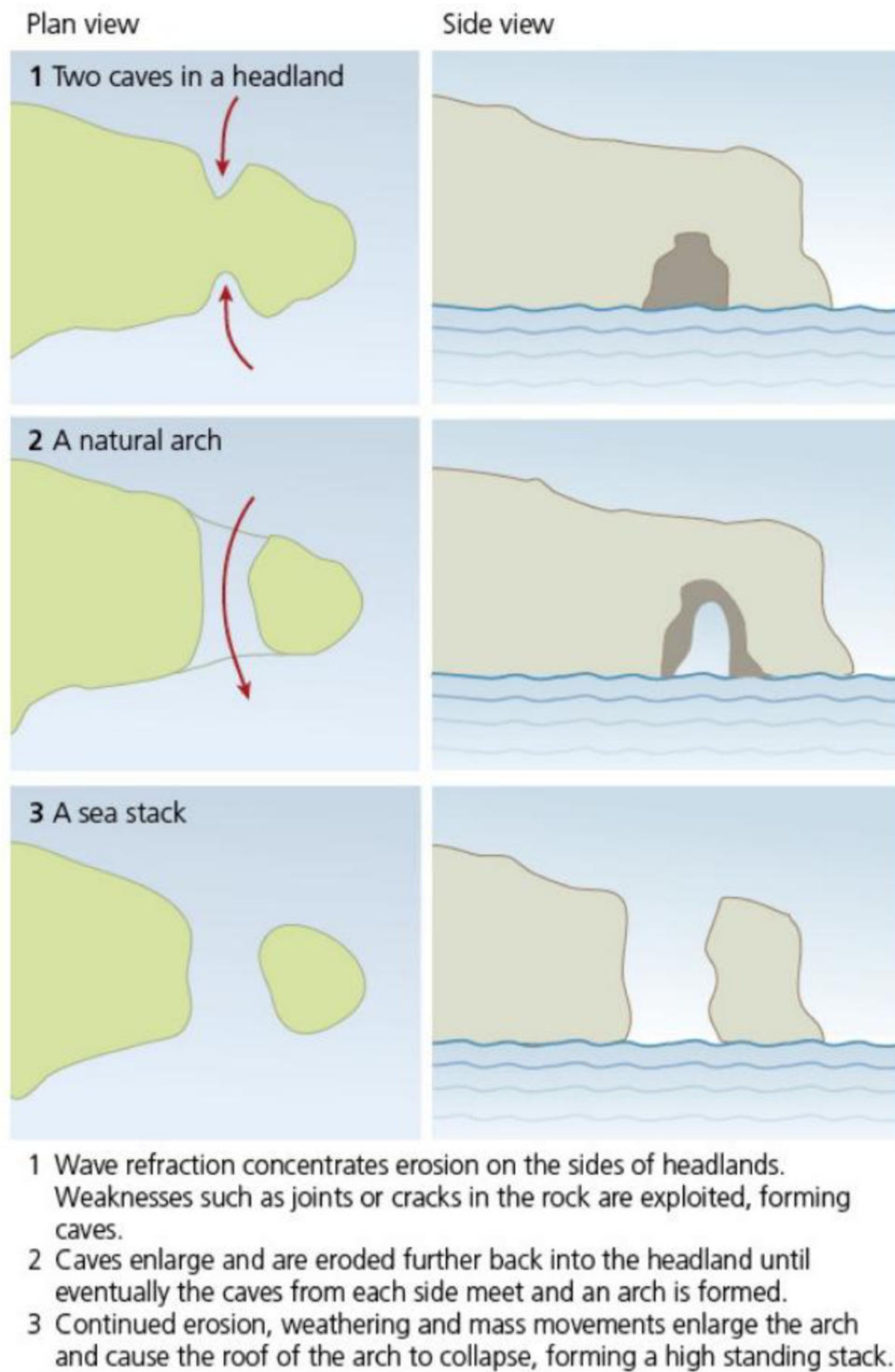


▲ **Figure 2.9** Headlands and bays



▲ **Figure 2.10** Headlands and bays in Praia de Rocha, Portugal

On a headland, erosion will exploit any weakness, creating, at first, a **cave** (Figure 2.11). Once the cave reaches both sides of the headland, an **arch** is formed (Figure 2.12). A collapse of the top of the arch forms a **stack**, and when the stack is eroded a **stump** is created (Figure 2.13). Where erosion opens up a vertical crack, allowing sea water to spout up at the surface, a blowhole is formed. The sandstone of the Cape Peninsula in South Africa has been attacked by the sea, forming steep vertical **cliffs** and small-scale features such as arches and stacks (Figure 2.14).



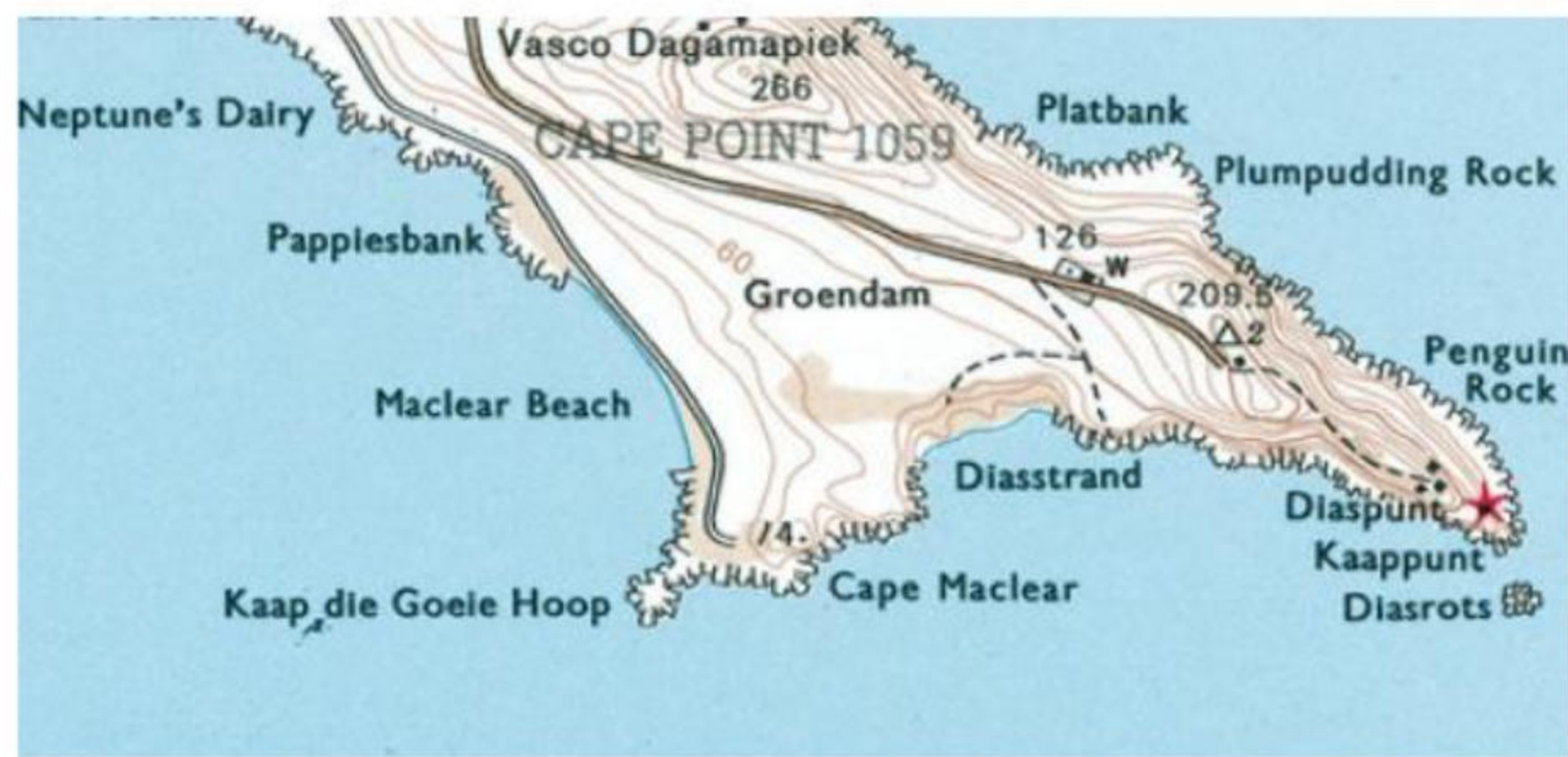
▲ **Figure 2.11** The formation of caves, arches and stacks



▲ **Figure 2.12** An arch at Durdle Door, Dorset, UK

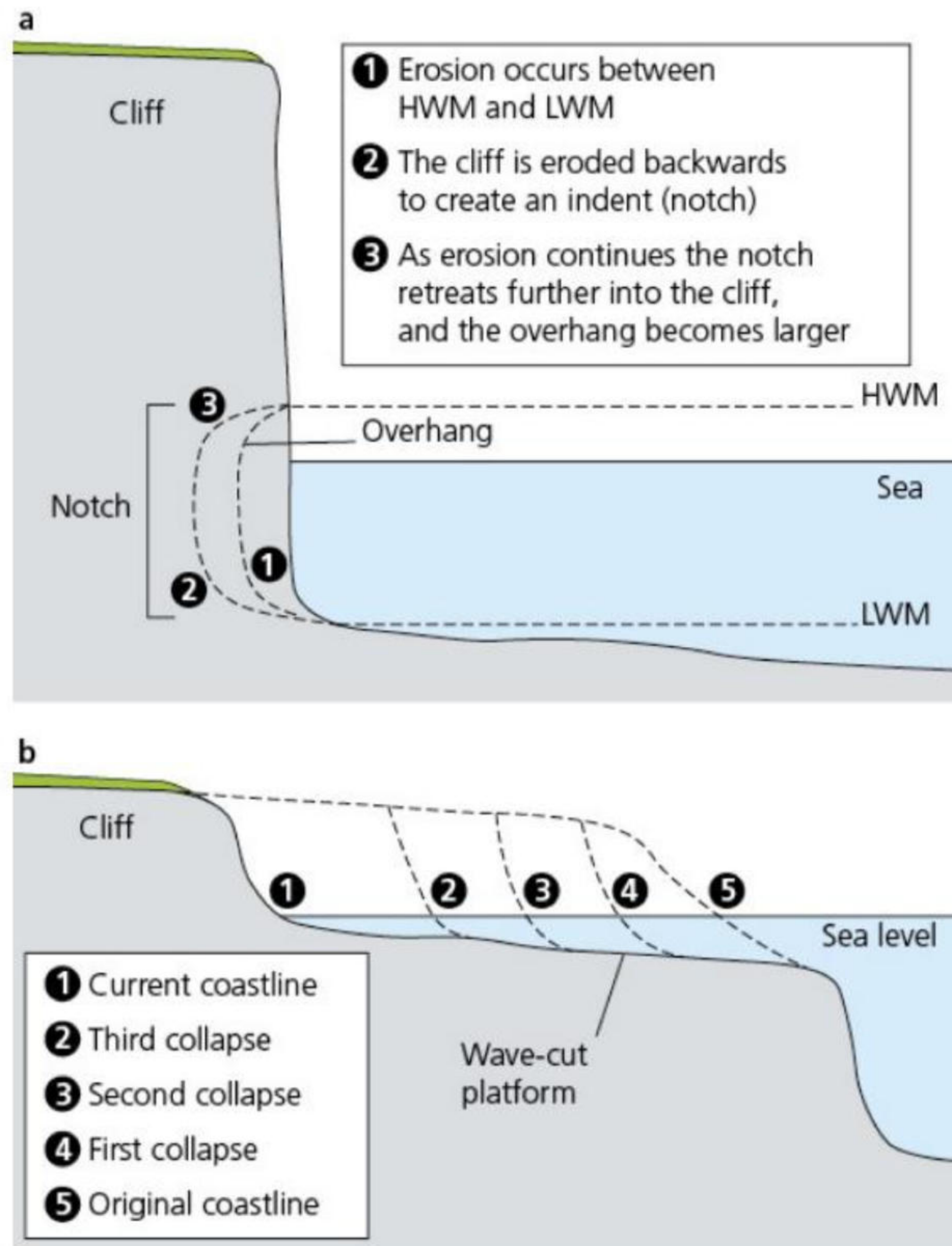


▲ **Figure 2.13** Stacks and stumps – part of the Twelve Apostles, Victoria, Australia



▲ **Figure 2.14** The Cape Peninsula, South Africa – showing Cape Maclear and the Cape of Good Hope in the background

Wave action is concentrated between the high water mark (HWM) and the low water mark (LWM). HWM is the level reached by the sea at high tide, while LWM is the level reached by the sea at low tide. The sea may undercut a cliff face, creating a notch and overhang ([Figure 2.15](#)). As erosion continues, the notch becomes deeper and eventually the overhang collapses, causing the cliff line to retreat. The base of the cliff is left behind as an increasingly longer platform. This is sometimes called a **wave-cut platform**, because it has been cut or eroded by wave action ([Figure 2.16](#)).



▲ **Figure 2.15** The formation of a wave-cut platform



▲ **Figure 2.16** A wave-cut platform, Cornwall, UK

Cliff profiles vary greatly, depending on the:

- » rate of coastal erosion (cliff retreat)
- » strength of the rock
- » presence of joints and bedding planes.

In addition, cliffs change over time, from ones that are dominated by marine processes (like wave erosion), to ones that are protected from marine processes but affected by land-based processes.

Activities

- 1 Explain the difference between attrition and corrasion (abrasion).

- 2 Explain why hydraulic action occurs in jointed rocks.
 - 3 State the type of rocks likely to be affected by corrosion (solution).
 - 4 Identify the types of erosion that are most likely to take place:
 - a during a storm
 - b on a beach
 - c on the face of a cliff.
 - 5 In your own words, describe how a wave-cut platform may be formed.
 - 6 Make a sketch of [Figure 2.10 \(page 39\)](#) and label the following features:
 - headland
 - bay
 - stack
 - beach.
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Beaches

The best beach development occurs on a lowland coast (constructive waves) with a sheltered aspect/trend, composed of 'soft' rocks that provide a good supply of material, or where longshore drift supplies abundant material.

On Tenerife, the lack of beach material other than volcanic material ([Figure 2.17](#)) has led to one beach, Las Teresitas, being formed of sand imported from the Sahara Desert ([Figure 2.18](#)). An artificial barrier prevents the sand from being eroded by wave action.



▲ **Figure 2.17** Volcanic beach, Las Teresitas, Tenerife



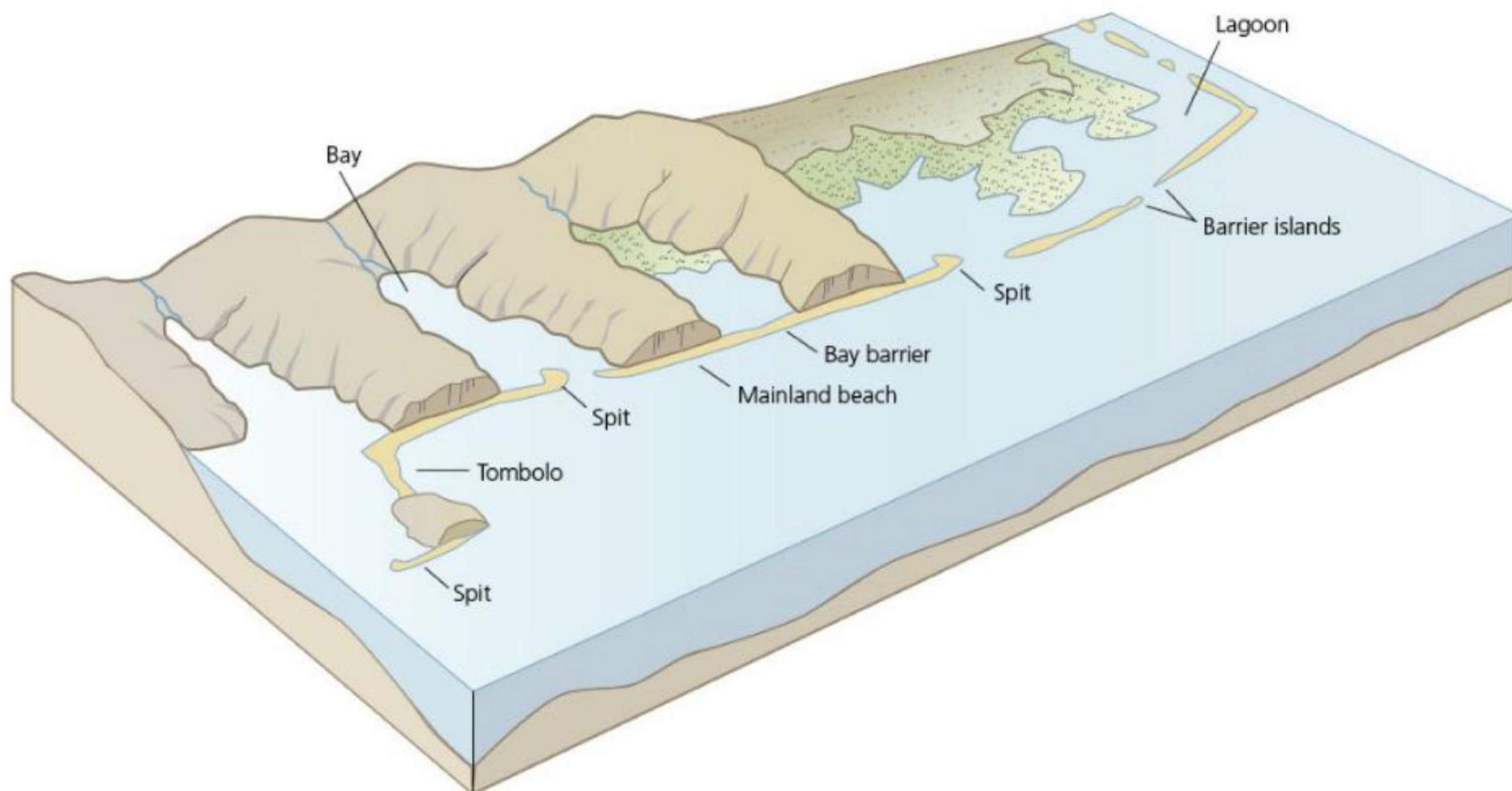
▲ **Figure 2.18** An artificial beach with imported sand

The term '**beach**' refers to the accumulation of material deposited between low spring tides and the highest point reached by storm waves at high spring tides. A typical beach will have three zones: backshore, foreshore and offshore. The backshore is marked by a line of dunes or a cliff. Above the high water mark there may be a berm or shingle ridge. This is coarse material pushed up the beach by spring

tides and aided by storm waves flinging material well above the level of the waves themselves. These are often referred to as storm beaches. The seaward edge of the berm is often eroded and irregular due to the creation of beach cusps.

The foreshore is exposed at low tide. The first material is deposited offshore. In this zone, the waves touch the sea bed and so the material is usually disturbed, sometimes being pushed up as offshore **bars**, when the offshore gradient is very shallow. Offshore bars are usually composed of coarse sand or shingle. Between the bar and shore, **lagoons** (often called sounds) develop ([Figure 2.19](#)). If the water in the lagoon is calm and fed by rivers, marshes and mudflats can be found. Bars can be driven onshore by storm winds and waves. A classic area is off the coast of the Carolinas in the southeast of the USA.

In 2017, a beach reappeared in Ireland that had disappeared (eroded) in storms in 1984 ([Figure 2.20](#)). Hundreds of tonnes of sand were deposited following freak tides, producing a 330 m stretch of golden sand.



▲ **Figure 2.19** The features of coastal deposition





▲ **Figure 2.20** The beach at Achill Island, Ireland that was washed away by storms in 1984 (top image) and returned after 33 years (bottom image)

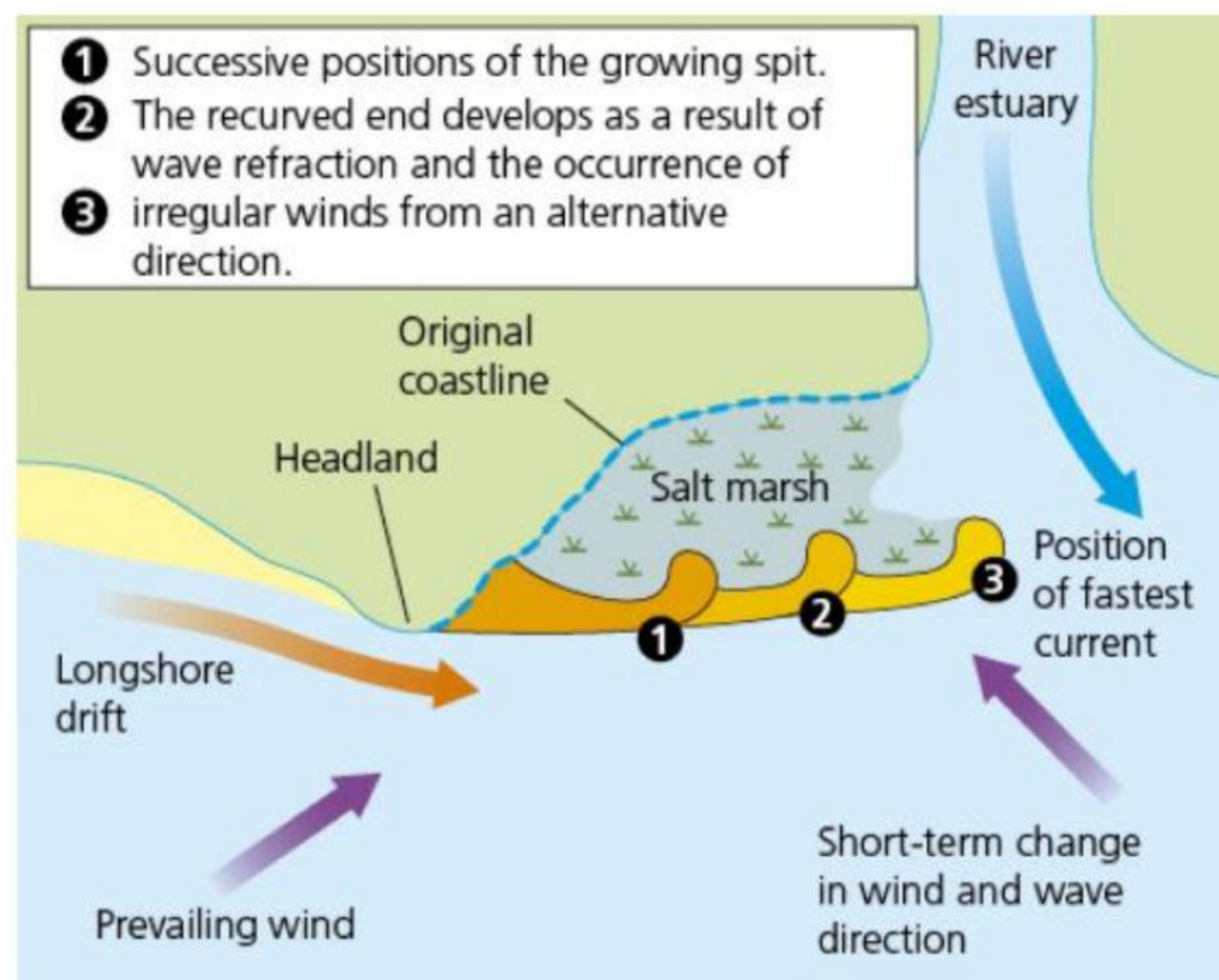
Bars and spits

Bars and spits are more localised features. They develop where:

- » abundant material is available, particularly shingle and sand
- » the coastline is irregular, for example where there is a variable geology
- » there are estuaries and major rivers.

A **spit** is a beach of sand or shingle that is linked at one end to land. It is found where wave energy is reduced, for example along a coast where headlands and bays are common and near river mouths (in estuaries and rias).

Spits often become curved as waves undergo refraction ([Figure 2.21](#)). Cross-currents or occasional storm waves may assist this hooked formation. A good example is the sandspit in Walvis Bay, Namibia. The main body of this spit is curved, but it has additional, smaller hooks, or recurves. Longshore drift moves sediment northwards along the coast. However, the coastline is very irregular here and there is a sudden change in the trend of the coastline. Consequently, refraction occurs, causing the waves to bend around eastwards.



▲ **Figure 2.21** The development of a spit

On the seaward side, the slope to deeper water is very steep. Within the curve of the spit, the water is shallow and a considerable area of mudflat and salt marsh is exposed at low water. These salt marshes continue to grow as mud is trapped by the marsh vegetation.

Related features include bars. These are ridges that block off a bay or river mouth ([Figure 2.22](#)). There are many examples on the west coast of Antigua ([Figure 2.23](#)).



▲ **Figure 2.22** A bar at Slapton Ley, Devon, UK



▲ **Figure 2.23** The west coast of Antigua. (Top) Mosquito Cove, (centre) Jolly Harbour Marina and (bottom) Crab Hill Bay

Interesting note

The longest spit in the world is the 112 km Arabat Spit in the Sea of Azov, between Russia and Ukraine. There are thirteen spits in the Sea of Azov.

Tombolos are ridges that link the mainland to an island. Good examples include the Lumley area of Sierra Leone, and the Cape Verde Peninsula, Senegal. The Cape Peninsula in South Africa is a complex tombolo that has developed on a very large scale.

The Palisadoes, Jamaica: a spit or a tombolo?

The Palisadoes is one of the largest deposited coastal features in the Caribbean ([Figure 2.24](#)).



▲ **Figure 2.24** The Palisadoes tombolo

Located just south of Kingston in Jamaica, this 13 km-long feature has been formed and re-formed many times during its history. Scientists believe that it may be 4000 years old.

Longshore drift occurs from east to west on the south coast of Jamaica. The sediment comes from rivers, cliff erosion and offshore sediments. The Palisadoes is located at a sharp bend in the coastline. Longshore drift carries sediment westwards and extends the length of the spit. As it grew longer it linked up with a number of cays (small islands), turning the spit into a tombolo.

The region experiences tropical storms and hurricanes. These can seriously damage the coast. For example, in 2004 Hurricane Ivan eroded up to a metre off the 2 m-high sand dunes. Even under normal conditions, summer sea breezes cause powerful destructive waves, which are capable of eroding the seaward face of the beach, causing it to become steeper.

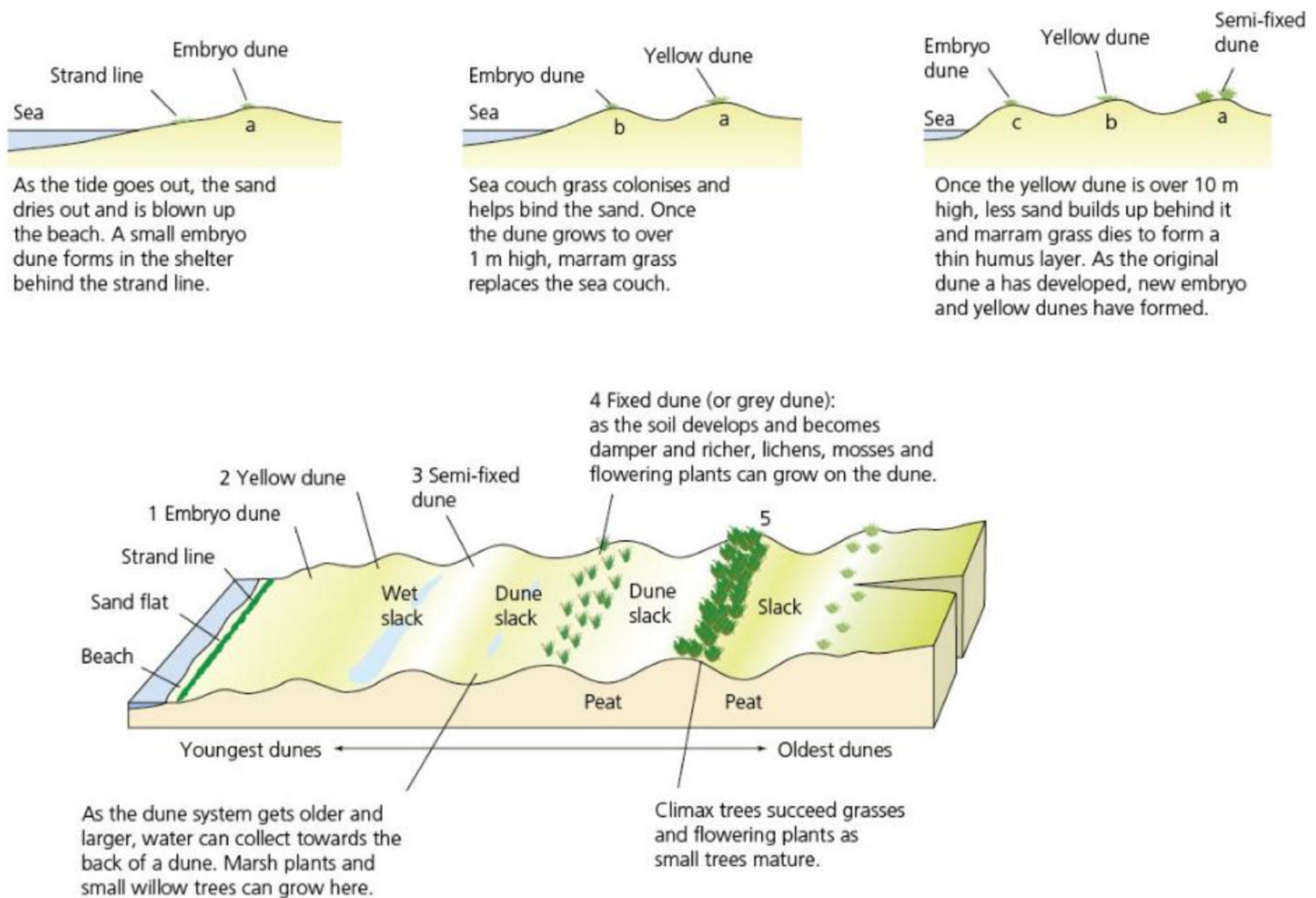
Activities

- 1 State:
 - a the age
 - b the length of the Palisadoes.
 - 2 Identify the direction in which longshore drift occurs on the Palisadoes.
 - 3 Identify where the sediment that builds up the Palisadoes comes from.
 - 4 Suggest the impact of hurricanes on the Palisadoes.
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Sand dunes

Sand dunes are one of the most dynamic environments in physical geography. Important changes take place here in a very short space of time. Extensive sandy beaches are almost always backed by sand dunes because strong onshore winds can easily transport the sand that has dried out and been exposed at low water. The sand grains are trapped and deposited against any obstacle on land to form dunes ([Figure 2.25](#)). Dunes can be blown inland and can therefore threaten coastal farmland and even villages. The interaction of winds and vegetation helps form sand dunes.

On the beach, conditions are very windy, dry (much water just soaks into the sand) and salty. Few plants can survive these extreme conditions but some can, including sea couch and marram grass ([Figure 2.26](#)). These are adapted to tolerate water with a high salt content and high wind speeds, and they can survive burial by sand. In fact, marram grass needs to be buried by fresh sand in order to send out fresh shoots.



▲ **Figure 2.25** The formation of sand dunes



▲ **Figure 2.26** Sand dune vegetation

Once marram grass and sea couch are established on a beach, they reduce the wind speed. This helps trap fresh sand. As the sand builds up, these plants send out new shoots, trapping more sand and building up a dune. Increasingly, the presence of plants in the sand dune adds organic matter and moisture to the dune and allows other plants to grow, such as heather.

The growth of new plants is called succession. Plants such as heather cannot tolerate the dry, windy, salty conditions of the beach, but they can survive in the less windy, moister, less salty dunes. They in turn alter the environment so that other species can invade and develop. Over a distance of just a few hundred metres on a sand dune, there may be as many as four or five different types of ecosystem.

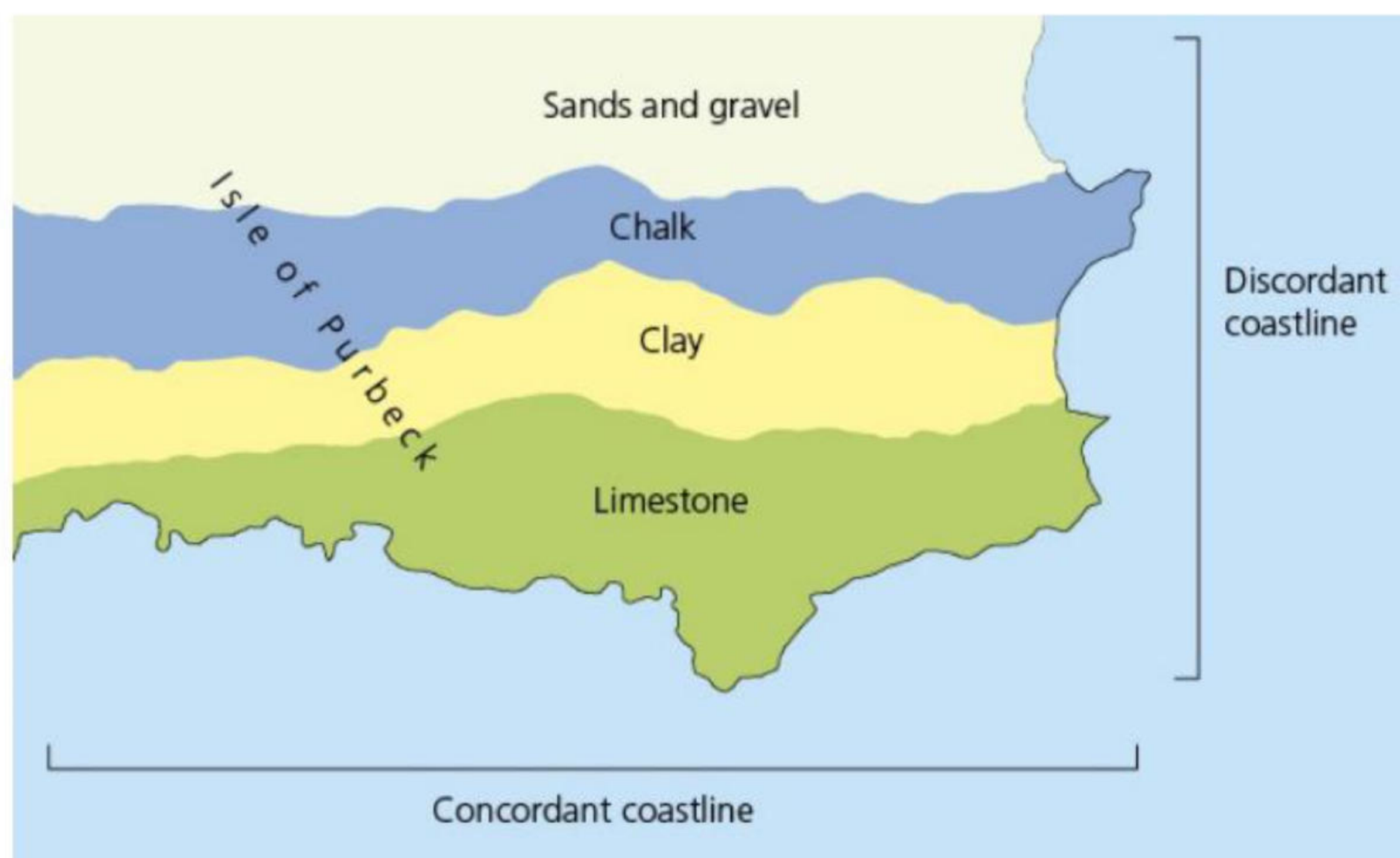
Activities

Study [Figure 2.25 \(page 45\)](#), which shows the development of vegetation on a sand dune.

- 1 Contrast the conditions between the shoreline and inland.
 - 2 Explain why deposition occurs on the sand dunes.
 - 3 Suggest how human activities might affect the sand dunes and/or the salt marsh.
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Discordant and concordant coastlines

Some coastlines have their geology parallel or at right angles to the shoreline. **Concordant coastlines** (also known as Pacific-type coastlines) tend to be straight and regular, with relief and geology parallel to the coastline ([Figure 2.27](#)), for example the south coast of the Isle of Purbeck or the coast of British Columbia. In contrast, **discordant coastlines** have relief and geology at right angles to the coast, for example the east coast of the Isle of Purbeck or northwest Spain.



▲ **Figure 2.27** Concordant and discordant coastlines